

Saumya Mathur

mathursaumya23@gmail.com | +91-70418-19901 | LinkedIn | GitHub

Computer Science undergraduate (VIT Vellore, 2026) with demonstrated full-stack engineering skills across the MERN stack, machine learning pipelines, and REST API design. Built three end-to-end projects spanning web applications, data science, and ML-based health monitoring. Seeking entry-level Software Engineer / Backend / Full-Stack Developer roles where strong fundamentals in DSA, system design, and modern web frameworks deliver immediate value.

EDUCATION

Vellore Institute of Technology, Vellore, India
Bachelor of Technology in Computer Science

Sep 2022 – May 2026
Graduation: 2026

PROJECTS

Team Task Manager — Full-Stack Web Application - Live Link

GitHub

MongoDB · Express.js · React.js · Node.js · JWT · Mongoose · REST API

- Architected a full-stack MERN project-management platform with JWT-based authentication and role-based access control (Admin / Member).
- Engineered 15+ RESTful API endpoints using Express.js and MongoDB/Mongoose to handle user management, project CRUD, task assignment, and progress tracking.
- Designed a React dashboard with dynamic task-status views (pending, in-progress, completed, overdue) for at-a-glance team productivity monitoring.
- Implemented secure route-protection middleware ensuring endpoints are accessible only to authenticated and authorised users.

Smart Community Health Monitoring — Hybrid ML System

GitHub

Python · Scikit-learn · XGBoost · LightGBM · Pandas · NumPy

- Built a weighted ensemble model (Random Forest + XGBoost + LightGBM) to predict water potability, achieving **93% classification accuracy** on real-world sensor data.
- Implemented Water Quality Index (WQI) computation, contamination classification, and disease-risk prediction (blue baby syndrome, fluorosis, bacterial contamination).
- Optimised performance through feature engineering, hyperparameter tuning, and ensemble weighting to minimise false negatives in health-critical predictions.

Crime Data Analysis — Unsupervised ML & Pattern Clustering

GitHub

Python · Pandas · NumPy · Scikit-learn · Matplotlib · PCA · K-Means

- Processed and standardised multi-year U.S. crime datasets; applied PCA for dimensionality reduction and K-Means clustering to surface long-term crime trend patterns.
- Evaluated cluster quality using Silhouette Score; produced labelled risk tiers (low / medium / high crime) to generate actionable policy insights from raw statistical data.

CERTIFICATIONS

- Java Developer — Introduction to Java, Java Intermediate
- MERN Stack Developer Certificate (View)
- Web Developer Certificate — Flixdin Training Camp (View)

TECHNICAL SKILLS

Languages	Java · JavaScript (ES6+) · Python · SQL · C++ · HTML5 · CSS3
Frontend	React.js · Responsive UI · Component Architecture · REST API Integration
Backend	Node.js · Express.js · REST API Design · JWT Authentication · RBAC
Databases	MongoDB · Mongoose ODM · SQL
ML / Data	Scikit-learn · XGBoost · LightGBM · Pandas · NumPy · Matplotlib · PCA · Ensemble Methods
Tools	Git · GitHub · VS Code · Postman
CS Fundamentals	DSA · OOP · Operating Systems · Computer Networks · DBMS